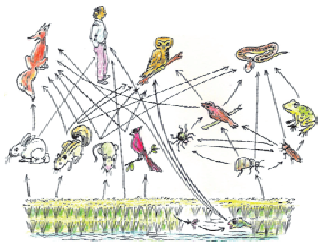
CHAPTER 5



OUR ENVIRONMENT: ENERGY FLOW IN THE ECOSYSTEM

You had studied in class 9 that the habitat of 'Kaani machri' is Kotumsar Caves. You had also studied about the habitats of several other organisms in the same chapter. Habitat is very important for the survival of organisms. The interrelationship of abiotic and biotic factors in the habitat of an organism play an important role in its life. We have studied in Chapter 1, Evolution that adaptation of organisms occur in different ways in habitats due to different abiotic and biotic conditions. The population of adapted organisms increases and impacts the population of other organisms of the same habitat. Come let us do an activity to study the interrelationship of organisms in their habitat.

5.1 Study of Interrelationships

Activity-1

Select an area around your house or school. It could be a part of a garden or a grassy field. The area should be such that you could observe it regularly (at least twice daily, both in the morning and evening) for 5 days. Mark a one meter by one meter area here. Observe carefully, do you find any insects, frog, bird or any other organism in the marked area? Write the name of the group of insects and the number of them that you observe there (as for example grass 20, ants 28 etc.). Dig the soil as well to count organisms. Try to keep counts such that you count a particular organism once only. If there are any trees, herbs, bushes etc. count them under the group of plants. Answer the following on the basis of your observations-

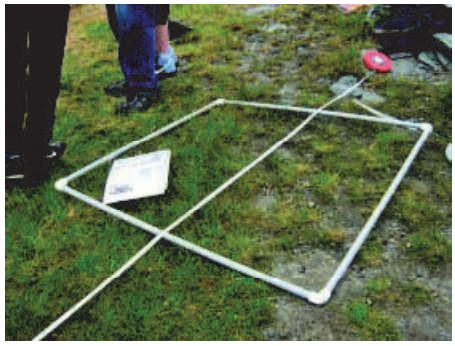


Figure-1: 1m X 1m Area

- Could you observe any food chain or food web in the marked area? Give an example.
- Could you observe any food web or food chain in the whole garden? Give an example.
- What are the biotic and abiotic factors in your area?
- What are the differences in abiotic factors of morning and evening hours?

Now calculate the average number of organisms observed per day in the area from the total organisms observed in 5 days(as for example if total number of ants were 50 those observed per day were 10 and this number multiplied by the area gives you and approximate number of ant population in that area)

- Do you think the organisms in the area have enough resources for their survival?

All organisms in an area depend on each other and their environment for some basic needs like food, reproduction and shelter.

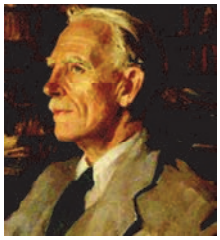
We can study this interrelationship of organisms and their environment on the basis of-

1. Interrelationship between biotic components- we shall study this with the help of food chains, food webs and ecological pyramids.
2. Interrelationship of biotic and abiotic components(the effect of environmental factors like soil, air, water, weather, climate etc. on organisms)- we shall study this with the help of nutrient cycle.

We shall try to study the flow of energy mainly among biotic components with a mention of about energy flow into the abiotic environment.

Food chain, food web, ecological pyramids, nutrient cycles are models that help us to study, present and understand the interrelationships of diverse organisms and their environment in either certain areas or in general. These help us to predict the environmental conditions and make comparative studies of different areas. We often use ways like that used in activity -1 to collect data regarding different parameters in an environment that we shall study as an ecosystem.

5.2 Meaning of an Ecosystem



A.G.Tansley

A self- coordinated systemdevelops in an environment, which we call as "Ecosystem", due to the interrelationship of environmental components. The word ecosystem was first used in the year 1935 by an environmentalist named A.G.Tansley. According to him the interrelationship of biotic and abiotic components in an environment forms a system of continuous exchange of nutrients energy etc.. By collecting, comparing and analyzing data collected of different parameters in the environment we can predict and find out the effect of certain specific parameters on the environment. Thus the study of the interrelationship based mainly on exchange of nutrients and energy between biotic and abiotic components is the study of an environmental system or in short ecosystem.

It is not by any means essential for conditions to remain same in a particular ecosystem. Conditions in an ecosystem undergo continuous changes due to diverse natural phenomena. Human interventions cause severe changes in environmental conditions in an ecosystem.

We can study the ecosystem of diverse areas of the environment like, a tree, garden, farm, forest etc. The habitat of Kani machri of Kotumsar caves would be an ecosystem study if we make an attempt to study the interrelationship of biotic and abiotic factors of the habitat.An example of a large ecosystem would be an ocean while a small one that of a cell. There are several microorganisms in a cell like bacteria, virus etc. that are present in the cellular environment.

Do you know?

Before the term ecosystem came in use a term called "Ecology" was widely in use. ErntHaekel had first used the term ecology to represent the study of nature as an oder similar to that of home. The literal meaning of the term 'Eco' in Greek is house and 'logos' is study. The definition of ecology was extended in the year 1869 to the study of mutual interdependence between plants and animals as well as with their environment. Ecology is thus a scientific study of identifying interrelationships among organisms along with that of their environment. Interrelationships could be related to food and feeding habits, reproductive behavior, parasitism, effect of environment on life cycles etc. These and several others have been studied over time to understand more about interrelationships.

Let us study an ecosystem to understand more about the interrelationships. We field plenty of rice farms in our state. Let us study the ecosystem of a rice farm.

5.2.1 Ecosystem of a rice farm

We know that most food chains start with plants. Plants are capable of converting solar energy to that stored in chemical bonds of compounds by the process called as photosynthesis. Plants are thus called as producers. You will study about this in detail in the chapter Life Processes Part-1. All organisms other than producers in a food chain are usually consumers. The producers in a rice farm are mainly the rice plants. Several organisms thrive in a rice farm from the sowing to harvesting season(and even thereafter). Some organisms thrive in the soil while some in the water while some others on the rice plant itself. There are occasional visitors as well that visit a rice farm sometime.We find an ecosystem changing over time on these rice farm. Let's study one-



Figure- 2: A rice farm

- You may have observed a rice farm, on the basis of your observation and the figure given here, enlist all the abiotic components in a rice farm.
- Figure 4 shows example of some organisms that thrive in a rice farm. Discuss with your friends and write the interrelationship between the organisms thriving in a rice field and the abiotic components there.
- What are the different food chains that you may observe in the given food web. Select and write any 5 of them.

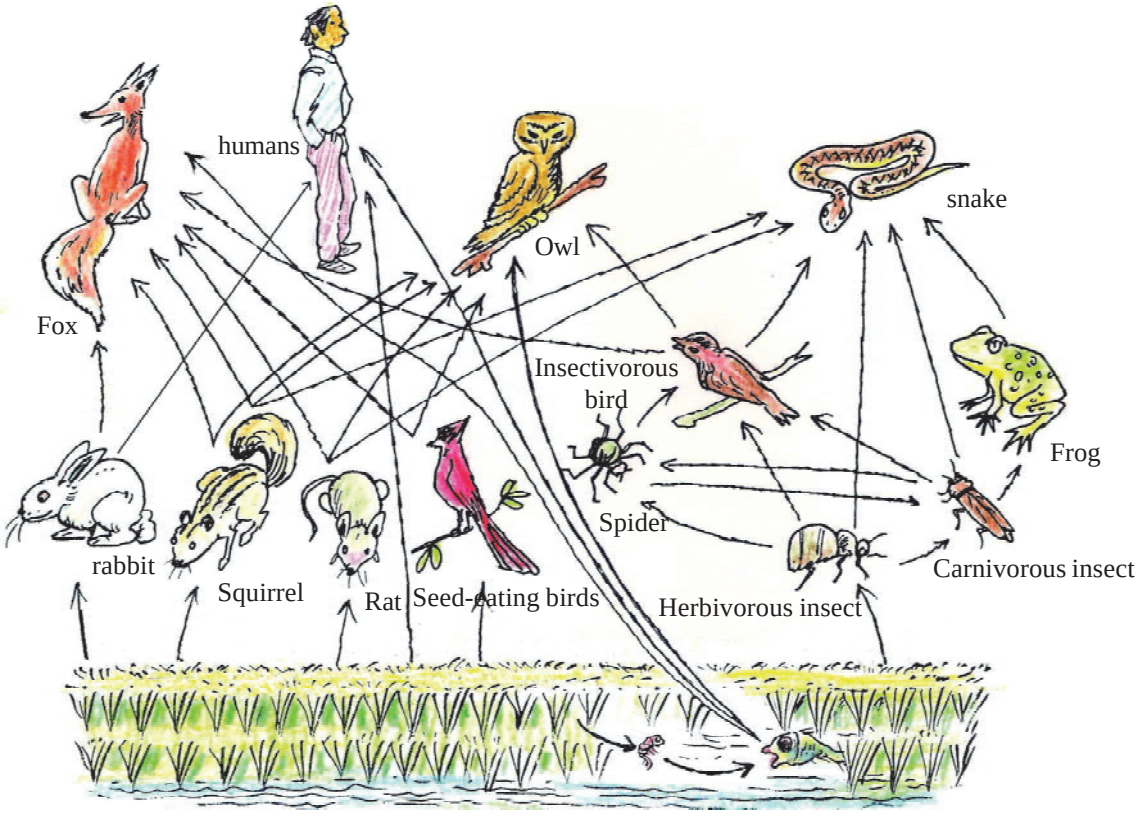


Figure-3: Food web in a rice farm

Studying a food chain of the food web in figure-4 and estimating the number of organisms found in a hectare(1000sqm) of a rice farm it was found that the numbers were in the following sequence-

100000 rice plants → 1000 rats → 50 snakes → 5 hawks.

Discuss with your friends about this example and answer the following-

- Which organisms are consumers in the given food chain?
- How many rice plants would feed a rat?
- A hawk feeds on how many snakes?
- It rats are killed how would it affect rice plants, snakes and hawks?
- If you answer the above question on the basis of the given food web, how do you think the killing of all rats would affect the population of the squirrels and the herbivorous insects (insect feeding on plants)?

There is such a lot of diversity in the living world that the relationship of eating and being eaten cannot be completely understood by a food chain. A food web gives a better preview of the same. We find as we study food chains and food webs that some organisms are solely dependent on plants while some others are solely dependent on other organisms. This gives us an idea that there are different levels in this relationship of eating and being eaten(those being eaten are called prey while those that eat them are predators). We call these levels as trophic levels. The rice plant at one trophic level while the rat another and the snake and hawk still others.

5.2.2 Ecosystem and Trophic levels

You may have observed several food chains in the food web of the rice field. We found there that the food chain started with mainly the rice plants. Those that feed on the rice plants like the sparrows, herbivorous insects, rats, rabbits, squirrels etc. are primary consumers.

- What are the organisms that feed on the primary consumers? These organisms are called secondary consumers.
- Are there organisms in the food web that consume the secondary consumers? Those that feed on secondary consumers are tertiary consumers.

Thus to study organisms feeding at different levels we consider them to be occupying a certain level of feeding called as trophic level that could be either producer or primary consumer or secondary consumer etc. as follows-

Producer	-	First Trophic level
Primary consumer	-	Second Trophic level
Secondary consumer	-	Third Trophic level
Tertiary consumer	-	Fourth Trophic level

Quaternary (usually also top) Consumer- Fifth Trophic level

Usually in a population of organisms of a particular trophic level we may find competitive feeding behavior. Those on a higher trophic level are dependent on the lower trophic level(second level would be dependent on the first). The population of the organisms of a trophic level is controlled by the population of the next or higher trophic level.

There are certain other organisms in nature that either feed on dead or decaying matter or on the wastes of other organisms. These may be scavengers like crows, cockroaches etc. or decomposers like bacteria, fungi etc.

- Which trophic level should these organisms belong to?
- Omnivorous organisms that feed on plants as well as animals, like human beings should be kept at which trophic level?

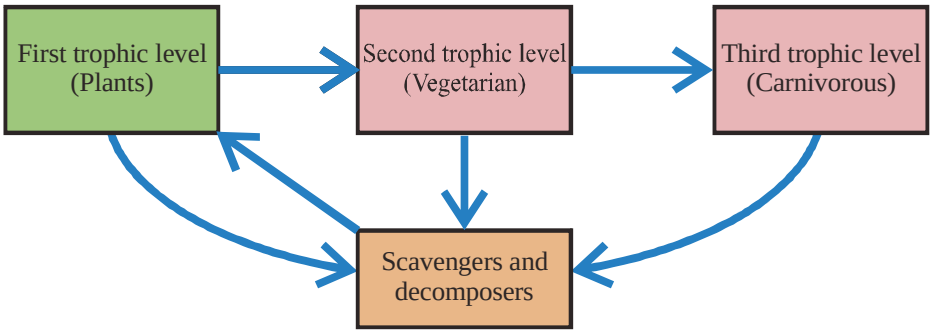


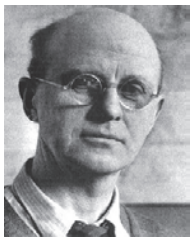
Figure-4: Trophic levels and decomposers

We can observe in figure5 that decomposers help in decomposition at all the trophic levels and help in providing the decomposed matter in the form of nutrient for producers at the first trophic level.

We shall use another model, the ecological pyramids to try to make a quantitative study of the interdependence between organisms, distribution of resources, flow of energy. It becomes easy to make a comparative study of different ecosystems with the help of ecological pyramids.

5.3 Study of Ecological Pyramids

5.3.1 Pyramid of Numbers



Charles Elton (1900-1991)

Charles Elton, an environmentalist wrote in his book "Animal Ecology" in the year 1927 for the first time about ecological pyramids. He discussed about ecological pyramid of numbers stating that in any area organisms that are present at the end of a food chain are those that produce least number of offspring (babies) in a year. Such an area is usually a habitat for the family of such an organism. Thus the feeding relationship of such an area can be represented in the form of a pyramid. The broad base of a pyramid would be occupied by organisms whose population would be maximum and the top would be with that of those with minimum population. Such a pyramid would represent the availability of food and the number of organisms at each trophic level. We do not find the diagram of ecological pyramids along the description given by Elton in his book. But we find that pyramids have been used extensively after Elton to not only represent numbers but other quantifiable parameters of ecosystems as biomass and energy. In most pyramids drawn so far quantified values have not been represented proportionately.

Do you know?

You may have heard about pyramids of Egypt. Pyramids are geometrical structures the base of which are broad and the shape gradually tapers to the top being nearly a point at the top. Usually we find pyramids with a square base with four triangular faces.



The organisms of the food chain of the rice farm can be represented in the form of pyramid as under.

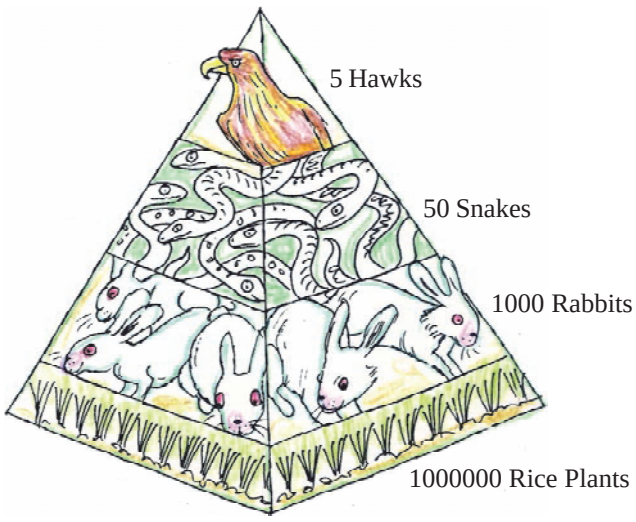


Figure-5 (a): Pyramid of number for organisms in a food chain (rice farm)

- Draw a pyramid to represent the number of organisms at each trophic level as obtained by you in Activity-1
- If we consider all organisms of a food web present at different trophic levels at a particular time the pyramid would look like this-

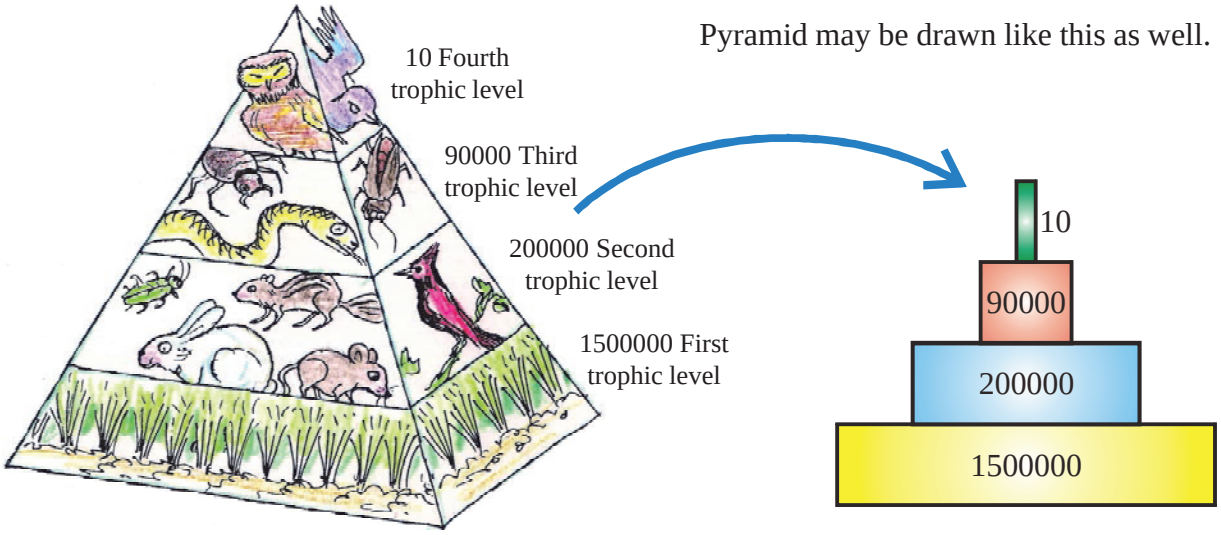


Figure-5 (b): Pyramid of number for organisms in a food web (rice farm)

- When the rice crop is harvested, what do you think will happen to the ecosystem on the farm?
- Do you think there would be the same number of organisms as shown in Figure-7 (b) still on the farm? What do you think will happen to them?

The condition of ecosystems are influenced and changed by human interventions and natural phenomena.

After harvesting several organisms die or move out to other areas where there are resources to sustain their survival.

It is thus often important to maintain certain conditions in an ecosystem for conservation of both biotic and abiotic types of resources.

You must have heard these days that it is recommended to grow more than one type of crops in a field at a time. Pulses are usually grown with cereal crops in many parts of our country. It is also recommended to grow trees in farm areas at particular intervals and also practice fish culture in the water that accumulates in the field. This has several benefits like conserving biodiversity, restoring soil nutrients, biologically controlling crop pests etc. We would need to study some other types of pyramids of numbers to understand these effects.

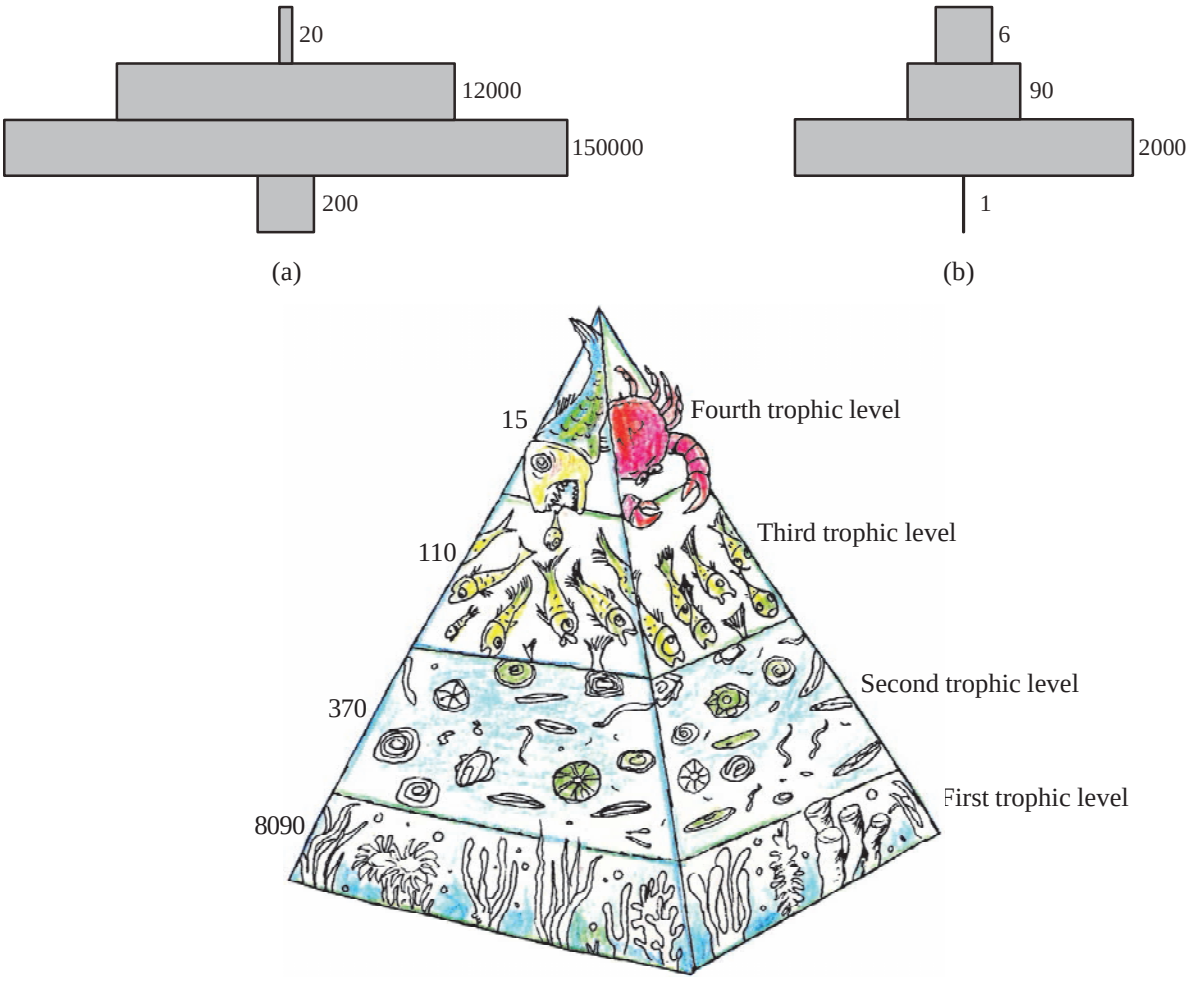


Figure-6: Pyramid of numbers (a) Forest (per hectare) (b) Tree (c) Pond (per hectare)

The ecological pyramid of numbers of a forest and that of a tree shows us that even if there are less organisms at the first trophic level, there are several at the next.

- Write a difference between the pyramid of numbers of the rice farm and that of a tree.
- Figure 8 (c) shows us a certain pyramid of numbers of an aquatic ecosystem. When will you find such relationships among aquatic organisms in a rice field?
- Write a similarity and a difference in the pyramid of numbers as shown in text of a rice field and that of an aquatic ecosystem like a pond.
- What does a broad base of a pyramid (as for example Figure 8 c) represent?
- Draw a food web according to any one of the above pyramids.
- How many trophic levels have been shown in each of the above examples?
- Calculate for each of the pyramids of figure 8 separately the number of organisms of first trophic level on which the next is dependent.
- If the number of organisms at the first trophic level or that of producers is less, will there still be enough food available for organisms of the next trophic level?

5.3.2 Role of Producers and Biomass

Primary production of a plant is the production of carbonic compounds by photosynthesis using solar energy and its accumulation. The sum total of the weight of plants(often by drying and taking dry weight) taken at a particular timerepresents the total primary productivity of that area. In most ecosystems, this is the biomass of the first trophic level or that of producers. Biomass may be calculated in this manner for the other trophic levels by estimating weight of organisms present at that level. A pyramid of numbers also indicates availability of food at each trophic level.Thus the availability of food at each trophic level does not necessarily depend on the pyramid of numbers but also on the pyramid of biomass.

- The pyramid of numbers of a forest and that of a rice farm are different. Do you think their pyramid of biomass would also be different?

5.3.3 Pyramid of Biomass

Examples of pyramid of biomass of some ecosystems is as follows-

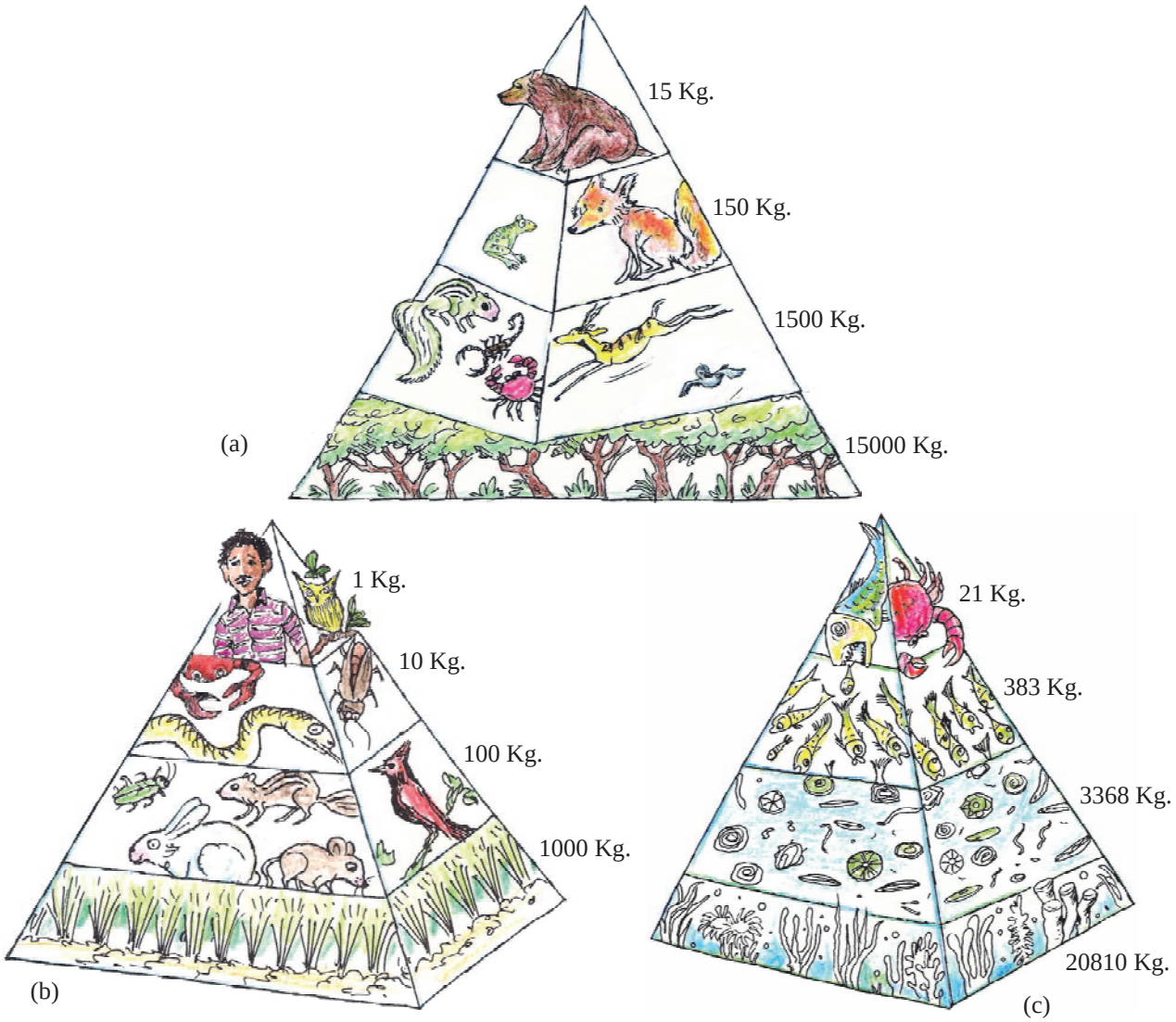


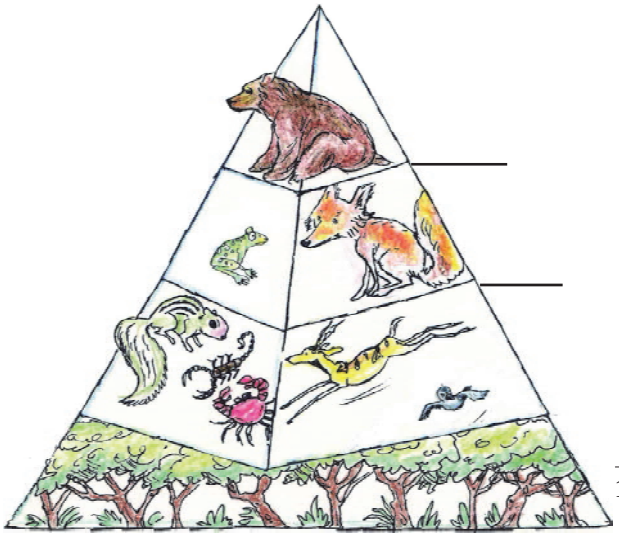
Figure-7: Pyramids of Biomass (Biomass in Kg/1000 square meter)
(a) Forest (b) Rice farm or a meadow (c) A type of aquatic ecosystem

- Can humans be kept at second or third trophic level in the pyramid of biomass shown in fig.9 (b)?
- Write any two similarities between the pyramids of biomass.
- Do organisms at the second trophic level have enough food according to the pyramid of biomass for a forest?

  Biomass is directly related to food and we know that we get energy from food.
- Observe the pyramid of biomass and guess if the energy obtained by the second trophic level is same as that of the third and fourth trophic levels.
- By what percentage did biomass reduce as we progressed from first to second trophic level?

5.3.4 Energy flow through trophic levels

It is very difficult to find out the amount of energy at each trophic level. We thus usually presume biomass as a basis to find out the energy at each trophic level. Thus let us find out the percentage of biomass available from first to next and further on to get an idea of the probable amount of energy that might also be available from one level to another (Note: though we usually follow this process of estimation of energy, there are several instances where even less biomass may produce energy equivalent to a lot of biomass).



$$\frac{1500}{15000} \times 100 = 10\%$$

Figure-8: Probable amount of energy available at each trophic level

Plants can convert nearly 1% of solar energy to that stored in bonds of chemical compoundssynthesized by the process of photosynthesis. Of this only 10% (0.1% of solar energy) is available for organisms of the next trophic level. Only a 10% of this again is converted to biomass at the second trophic level and available to those of the third trophic level. Thus efficiency of energy conversion at each trophic level is 10%. The efficiency of energy conversion in different ecosystems varies and values ranging from 2% to 24% have been calculated in researches conducted so far. Some amount of energy in plants is used to carry out its life processeswhile some is expended into the environment. Thus a very small amount of energy is available to the next trophic level. Similarly a much lesser amount of energy is available the third fourth and fifth trophic levels. Thus the base of all energy pyramids starting with producers at the first trophic level is broadest. Such pyramids are never inverted. This helps us to understand that no matter how varied ecosystems maybe, the energy flow in most of them occurs in a similar manner.

- How many trophic levels were there in the food chain that you observed in activity 1?
- What were the maximum number of trophic levels in the rice farm?

- How many trophic levels do we find in any food chain? Why do you think it is so?
- If in a rice farm there is 5000 kilocalories of energy at the first trophic level /or the level of producers, then how many trophic levels will be there till 0.5kilocalories of energy? Calculate and draw a pyramid to show energy at each trophic level(note that at each level the efficiency of energy conversion is 10%). Does the food chain of activity 1 have similar number of trophic levels?

Any food chain actually represents the pathway of flow of energy. A food web shows several such pathways. It becomes clear from food chain and pyramid of energy, that the flow of energy is unidirectional. The energy converted by plants cannot be converted back to solar energy. See figure 11 to understand more about energy flow through trophic levels.

Think and discuss

- Do you think energy in lost/ destroyed at each of the trophic levels?(remember you had studied in chapter 3 about conversion of energy from one form to another)
- What do you think is the role of decomposers in the flow of energy (see figure11)

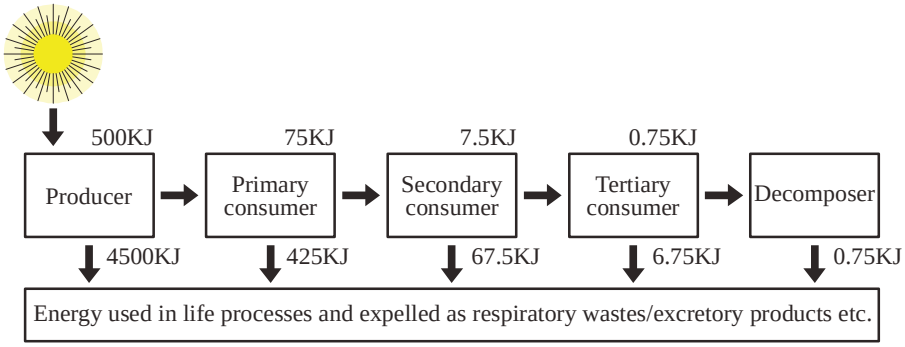


Figure-9: Estimated energy flow in an ecosystem

Do you know?

Food chains and food webs also start with decomposing substances. You had studied about one such pathway in the habitat of kani machri of Kotumsar caves. Here is another such example. Organisms that grow on a decomposing body of a bird like certain bacteria or yeasts are eaten by paramecium and spider feeds on insects that feed on decomposing wastes.



Decomposing substance — primary consumer — secondary consumer

Source of energy

At the bottom of oceans where solar energy does not reach, there are food webs that start with bacteria that thrive not on solar energy but energy in the chemicals of the hot springs. These hot springs arise at the vents of volcanoes at the ocean floor and the temperature is often as hot as 400°C.

We observed that at each trophic level energy is expelled or expended through wastes or in running life processes respectively. Wastes produced contain energy that are no longer of use to the organism that produces them. They can be useful to others and thus substances in nature are constantly being recycled. Chemical compounds in abiotic substances in nature like wastes of living organisms and other chemicals present in soil, air and water are in constant circulation from a biotic to abiotic source and vice versa. We may study about this in the form of cycle of nutrients.

5.4 Nutrient cycles- flow of substances

You know that there are over 100 elements on earth. These elements are either in solid, liquid or gaseous states in nature. Bodies of living organisms are formed by some of these elements. Several organic compounds like carbohydrates, proteins, fats, nucleic acids etc. are synthesized in living bodies which are all carbon compounds. Often we study about these as carbon, nitrogen or other cycles.

Think and note

- Where do we find carbon dioxide gas?
- What are the different sources of carbon dioxide?
- Through which life process carbon compounds are synthesized in plants?
- How do these compounds reach the body of animals?

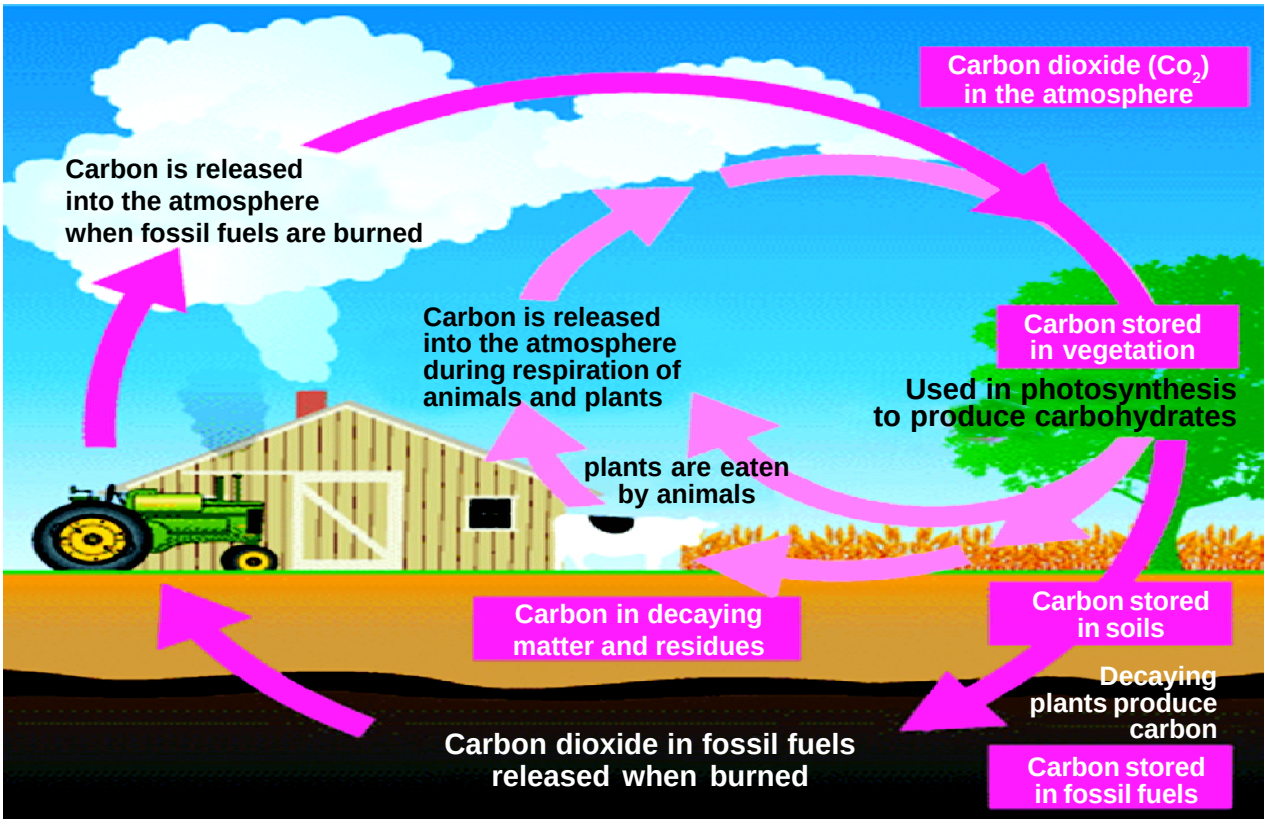


Figure-10: Carbon cycle

- Observe the figure and find out the sources from which carbon dioxide reaches the atmosphere?

Usually carbon cycles mainly in the form of a gaseous cycle of carbon dioxide. Green plants use atmospheric carbon dioxide to synthesize several carbon compounds. These are used and certain other carbon compounds(mainly proteins and fats) are often synthesized by consumers. The combustion of these compounds by different processes sends carbon-dioxide (CO_2) back into the atmosphere. The process of respiration (also a process of combustion) in living organisms is one such process. The process of decomposition also releases carbon dioxide into the atmosphere. Other simple carbon compounds formed also enter the nutrient cycle of nature.

5.5 Human intervention in ecosystems

Human beings impact ecosystems in several ways. We may see in the cabon cycle itself that the excess use of fossil fuels increases the level of oxides of carbon and carbon particles (both being potent sources of pollution).

Think, discuss and write

- Which abiotic components of the atmosphere do you think have been adversely affected by human intervention?
- Which biotic components have been adversely affected by humans?

Let us take the example of a rice farm, humans for their own benefits, destroy most organisms of the second trophic level.

- What will be the effect of this on the ecosystem of the rice farm?
- How will the impact of introduction of a new species of organism into a food chain in a rice farm?
- If 0.01milligram of arsenic accumulates in a rice plant due to use of arsenic containing pesticides, what will the total amount of accumulation in 30 plants? If you feed on grain obtained from around 30 plants everyday, think how much arsenic reaches your body every day?

We know that to protect our farms and get maximum produce we use different types of chemicals. Several of these like herbicides, insecticides, fungicides etc. are used to remove unwanted plants growing in farms, insects and disease causing macro and microorganisms. Some of these chemicals are decomposed quickly while several others remain in the soil for long periods of time as such and adversely affect crop plants, beneficial organisms etc. along with the harmful organisms. Sometimes they are absorbed into crop plants and affect humans and other organisms feeding on them adversely by causing several harmful diseases.

Chemicals that remain in the soil often affect the physical and chemical properties of the soil turning it unfit for our crops. This reduces crop production over time. Conversion of such areas into fertile land is very difficult, thus it is suggested that use of artificially manufactured chemicals be reduced to protect both our environment and us.

- Write about some ways in which observe around you how humans adversely affect the environment.

- What could be your role to save the environment from any harm(even if it is just at home or in your locality)? Write in detail about it.

Do you know?

Water hyacinth is a fast growing water plant. You may have observed these spread over the surface of in ponds and other water bodies of your locality. This plant was imported from America to our country. The introduction of this new species mainly for ornamental purpose has adversely affected both biotic and abiotic components of several water bodies.



What we have learnt

- All organisms in nature are interdependent. They also depend on the nonliving or abiotic components of the environment.
- Ecosystem is a system of interdependence for energy among biotic and abiotic components of nature.
- Most plants convert solar energy to energy conserved in bonds of chemical compounds by the process of photosynthesis and thus are called as producers.
- Primary productivity is the rate at which energy is converted by mainly photosynthetic plants to organic substances. The total amount of productivity in a region or ecosystem is total primary productivity.
- Total primary productivity of a given area also gives us the biomass of producers of a given area(it may be determined by taking the weight of producers in the area).
- All organisms other than producers are mainly consumers in a food chain.
- In the food and feeding relationships as observed in food chains or food webs, each level starting mainly from producers is called as trophic levels.
- Number of organisms, biomass or energy at each trophic level can be represented by drawing pyramids of number, biomass and energy.
- Biomass may be estimated by weight of organisms of a trophic level of a certain area at a certain time.
- Plants can convert nearly 1% of solar energy to that stored in bonds of chemical compounds by the process of photosynthesis of which only 10% is available for organisms of the next trophic level. Only a 10% of this again is converted to biomass at the second trophic level and available to those of the third trophic level. Thus the efficiency of energy conversion at each level is around 10%.
- Nutrients are constantly cycled in nature between biotic and abiotic components.

Key words

Ecology, ecosystem, trophic level, ecological pyramid, biomass, nutrient cycle

Exercise

1. Choose the right option.

(i) There are about 2234 insects, 56 birds and 3 snakes on a mango tree. The pyramid of energy would be-

(a) Straight (b) Inverted (c) rectangular (d) uncertain

(ii) Fungi started growing on a moist piece of bread left in the open. After sometime some insects like houseflies etc. were seen there. The organisms at the last trophic level would be-

(a) Food (b) bread (c) house fly (d) none

(iii) There are 3 stages in the life cycle of an insect - egg, larva, pupa, adult. If the stages of egg, larva and pupa are completed in the body of a particular organism then the insects life cycle is completed in how many ecosystems?

(a) 1 (b) 2 (c) 3 (d) 4

(iv) What will happen if all insect eating birds were removed from a particular cropland area?

(a) Crop production will increase (b) Insect infestation will increase

(c) Increase in the number of other birds (d) No effect

(v) Cow feeds on grass. Its dung is used to make dung cakes which are used as fuel. The burning of the dung cakes produce smoke that reaches the atmosphere and the ashes are mixed with soil. This adds nutrients for grass to grow. This process is an example of-

(a) Food chain (b) Food web (c) Nutrient cycle (d) Life cycle

(vi) The food chain in a pond starts from some water plants that are eaten by certain fishes those eaten by others extending finally to humans. The energy here from one trophic level to another will-

(a) Gradually decrease (b) gradually increase

(c) will be same (d) will be less sometime and more sometime.

(vii) The number of organisms in a garden was estimated as 5567 grasses, 453 shrubs, 23 trees and 7769 animals. The primary productivity of this area will be nearly equal to-

(a) Biomass of all organisms of the garden

(b) Biomass of all the plants of the garden

(c) Biomass of the animals of the garden

(d) Biomass of only 5567 grasses

2. A type of fungi is grown on rice husk. We eat the fungi. Does it show a food chain. Illustrate the food chain. What is the source of energy for this food chain?
3. How does energy flow in any ecosystem? Explain in your own words.
4. What will happen if only plants and humans are left on earth as living organisms or biotic component?
5. What will happen if the natural flow of energy in an ecosystem is disrupted? Explain with an example.
6. How does the availability of energy affect the number of organisms at different trophic levels?
7. Site an example where you find the effect of human beings on an ecosystem. Explain why you think so and suggest some means to minimize the same.

Note: The numerical values given in most of the illustrations have been provided only for study and are not meant to be learnt by rote. Also note that ecosystems are so varied that values of such type cannot apply universally. Even two ponds would not have same values.